

AP Vidyut Recruitment Exam 23rd to 27th Aug 2026

Hall Ticket Number	APVR0016548
Participant Name	KRISHNA NAGA VENKATA SAI TEJA
Test Center Name	Shree Vijaya Durga I.T Info solutions
Test Time	2:30 PM - 5:30 PM
Test Date	26/08/2026
Subject	Electrical Engineering

Section : General Intelligence and Reasoning

- Q.1 Refer to the following number, symbol series and answer the question. Counting to be done from left to right only. (Note: All numbers are single digit numbers only)
(Left) 2 * £ 5 7 Ω 1 3 8 # \$ 9 4 % 5 @ 6 ^ (Right)
How many such symbols are there each of which is immediately preceded by an odd number but not immediately followed by an even number?¹⁰¹⁴³

Ans ☒ 1. One
☐ 2. None
☐ 3. Two
☐ 4. More than two

Question ID : 441009796127

- Q.2 Which of the following letter-clusters should replace # and % so that the pattern and relationship followed between the letter-cluster pair on the left side of :: is the same as that on the right side of :: ?

: DYR :: TOH : %

Ans ☒ 1. # = MHA, % = KFY
☐ 2. # = MHB, % = KFY
☐ 3. # = MHA, % = KFI
☐ 4. # = NHA, % = KFY

Question ID : 441009786346

- Q.3 GWXI is related to KABM in a certain way based on the English alphabetical order. In the same way, DTFQ is related to HXJU. To which of the following is LYSE related to, following the same logic?⁶⁰⁶³

Ans ☒ 1. PCWI
☐ 2. MZTF
☐ 3. PZXF
☐ 4. MCWF

Question ID : 441009818604

Q.4 75 is related to 29 following a certain logic. Following the same logic, 66 is related to 26. To which of the following is 30 related to, following the same logic?
(NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)

- Ans
- ☐ 1. 13
 - ☐ 2. 15
 - ☐ 3. 12
 - ☒ 4. 14

Question ID : 441009585206

Q.5 Seven people, A, B, C, D, E, F and G are sitting in a row, facing north. Only two people sit to the left of C. Only three people sit between E and D, neither of whom sits on the extreme end. A sits immediately to the right of F. G sits immediately to the left of E. Who sits third to the left of D?

- Ans
- ☐ 1. G
 - ☒ 2. C
 - ☐ 3. A
 - ☐ 4. E

Question ID : 441009814181

Section : General awareness

Q.6 The Inter Tropical Convergence Zone (ITCZ), sometimes referred to as the monsoon trough over the Gangetic plain during July, had been characterised by which of the following atmospheric conditions?

- Ans
- ☐ 1. A high-pressure belt associated with descending dry air currents
 - ☐ 2. A cold air front formed due to polar continental air masses
 - ☒ 3. A low-pressure zone where trade winds converge and air tends to ascend
 - ☐ 4. A permanent subtropical jet stream over north-western India

Question ID : 4410091800823

Q.7 Who was the prominent leader of the Wahabi Movement in India that integrated religious revivalism with socio-economic resistance?

- Ans
- ☐ 1. Birsa Munda
 - ☒ 2. Syed Ahmed Barelvi
 - ☐ 3. Shariatullah Khan
 - ☐ 4. Dadu Mian

Question ID : 441009865974

Q.8 The Surat Session of 1907 is remembered for a major split in the Indian National Congress. Which of the following was the main consequence of this session for the national movement?

- Ans
- ☐ 1. Proposal for Home Rule League
 - ☐ 2. Adoption of Non-Cooperation Movement
 - ☐ 3. Formation of the Muslim League
 - ☒ 4. Separation of Moderates and Extremists

Question ID : 4410092115972

Q.9 Amid ongoing regional conflicts in 2026, reports indicate that the Chabahar Port developed and operated by India has sustained damage. In which country is this strategic port located?

- Ans
- ☐ 1. Oman
 - ☐ 2. Iraq
 - ☐ 3. Egypt
 - ☒ 4. Iran

Question ID : 4410091496380

Q.10 Which one of the following languages was conferred the Classical Languages status in 2024?

- Ans
- ☐ 1. Sanskrit
 - ☐ 2. Kannada
 - ☐ 3. Tamil
 - ☒ 4. Marathi

Question ID : 441009936117

Q.11 Which of the following statements is INCORRECT with reference to Kamarajar Port?

- Ans
- ☐ 1. It is the first port in India to become a public company.
 - ☒ 2. It is one of the Minor Ports in India.
 - ☐ 3. It is located on the Coromandel Coast.
 - ☐ 4. It is located in the state of Tamil Nadu.

Question ID : 44100979775

Q.12 Which factor contributes most to the arid climate of the Thar Desert region in northwestern India?

- Ans ☒ 1. Frequent tracks of intense tropical cyclones originating over the Arabian Sea
- ☒ 2. High surface albedo and extreme soil salinity characteristic of the northern plains
- ☒ 3. The parallel alignment of the Aravalli Range to the direction of the southwest monsoon winds
- ☒ 4. Perennial flooding and high water tables of the Indus river tributaries

Question ID : 4410092258040

Section : English

Comprehension:

Read the passage carefully and answer the questions that follow.

The global energy transition is increasingly shaped not only by the expansion of renewable power generation but also by advances in battery storage technologies capable of addressing the intermittency of solar and wind energy. Recognising this, governments are supporting large-scale battery manufacturing, grid-scale storage systems, and research into alternative chemistries that reduce dependence on scarce critical minerals. However, experts caution that storage capacity alone cannot guarantee energy security unless transmission infrastructure, recycling mechanisms, and regulatory frameworks evolve simultaneously. Consequently, recent policy initiatives emphasise integrated energy ecosystems rather than isolated technological achievements. This perspective suggests that the durability of the clean-energy transition will depend less on individual innovations than on the coordinated development of complementary industrial, environmental, and institutional capabilities across diverse economies worldwide.

SubQuestion No : 13

Q.13 Which of the following can be most reasonably inferred from the passage?

- Ans ☒ 1. Dependence upon critical minerals will inevitably disappear because alternative battery chemistries are expected to replace conventional technologies entirely.
- ☒ 2. Renewable-energy technologies have largely eliminated the operational challenges historically associated with electricity transmission and storage infrastructure worldwide.
- ☒ 3. Expanding battery production without corresponding institutional and infrastructural reforms is unlikely to ensure a resilient clean-energy transition.
- ☒ 4. Government investment in battery manufacturing has become unnecessary since private-sector innovation sufficiently addresses long-term energy-security requirements independently.

Question ID : 4410092414154

Comprehension:

Read the passage carefully and answer the questions that follow.

The global energy transition is increasingly shaped not only by the expansion of renewable power generation but also by advances in battery storage technologies capable of addressing the intermittency of solar and wind energy. Recognising this, governments are supporting large-scale battery manufacturing, grid-scale storage systems, and research into alternative chemistries that reduce dependence on scarce critical minerals. However, experts caution that storage capacity alone cannot guarantee energy security unless transmission infrastructure, recycling mechanisms, and regulatory frameworks evolve simultaneously. Consequently, recent policy initiatives emphasise integrated energy ecosystems rather than isolated technological achievements. This perspective suggests that the durability of the clean-energy transition will depend less on individual innovations than on the coordinated development of complementary industrial, environmental, and institutional capabilities across diverse economies worldwide.

SubQuestion No : 14

Q.14 Which of the following best describes the organisational structure of the passage?

- Ans ☒ 1. It identifies a contemporary challenge, explains an emerging policy response, introduces a qualifying limitation, and concludes with a broader strategic implication.
- ☐ 2. It begins with technological achievements, critiques environmental regulations comprehensively, and concludes by questioning the feasibility of renewable-energy adoption altogether.
- ☐ 3. It presents contrasting economic theories regarding renewable investment, evaluates supporting statistical evidence, and ultimately endorses one theoretical perspective exclusively.
- ☐ 4. It traces the historical evolution of battery technologies, compares competing storage chemistries, and predicts their future commercial dominance across global markets.

Question ID : 4410092414155

Q.15 Change the given sentence into passive voice.

Inform the mother about your result.²¹⁴³

- Ans ☐ 1. Let the mother being informed about your result.
- ☐ 2. Let the mother informed about your result.
- ☒ 3. Let the mother be informed about your result.
- ☐ 4. Let the mother be inform about your result.

Question ID : 441009497397

Q.16 Complete the following sentence by choosing the correct option.

The professor asked the students to ____ the main arguments presented in the article and compare them to current research.

- Ans ☐ 1. neglect
- ☒ 2. synthesize
- ☐ 3. imitate
- ☐ 4. contradict

Question ID : 4410091711934

Q.17 Choose the most appropriate one-word substitute for the word in bold:
The **pure and peaceful** environment of the cemetery was shattered by the loud honking of the cars.

- Ans
- ☒ 1. seraphic
 - ☐ 2. lustre
 - ☐ 3. embers
 - ☐ 4. aurora

Question ID : 4410091538212

Section : Quantitative aptitude

Q.18 Find the simple interest (in ₹) if a sum of ₹500 is borrowed for 7.5 years at 20% per annum rate of interest.776

- Ans
- ☐ 1. 850
 - ☒ 2. 750
 - ☐ 3. 800
 - ☐ 4. 730

Question ID : 441009852058

Q.19 Three friends P, Q and R started a business by investing a sum of money in the ratio of 4 : 9 : 8. After 6 months Q withdraws half of his capital. If the sum invested by P is ₹80000, out of a total annual profit of ₹15000 find the share of Q in profit.9698

- Ans
- ☐ 1. ₹3995
 - ☒ 2. ₹5400
 - ☐ 3. ₹5705
 - ☐ 4. ₹5635

Question ID : 441009548025

Q.20 Evaluate: $25 \div (5 \times 5)$ 5415

- Ans
- ☐ 1. 3
 - ☐ 2. 4
 - ☒ 3. 1
 - ☐ 4. 2

Question ID : 441009521699

Q.21 What is the difference between the average of prime numbers between 50 and 60 and the average of multiples of 5 between 20 and 50?4036

Ans ☒ 1. 21

☐ 2. 29

☐ 3. 14

☐ 4. 27

Question ID : 441009580940

Q.22 A car covers a distance of 1890 km in 14 hours. Speed of a train is double the speed of the car. One-third the speed of the train is equal to the speed of a bike. How much distance will the bike cover in 5 hours?10944

Ans ☐ 1. 453 km

☐ 2. 449 km

☒ 3. 450 km

☐ 4. 458 km

Question ID : 441009549271

Q.23 The population of a village was 320000. It increased by 5% in the first year and increased by 20% in the second year. Its population after two years is _____.7652

Ans ☒ 1. 403200

☐ 2. 384000

☐ 3. 336000

☐ 4. 400000

Question ID : 441009858934

Q.24 Provisions meant for 104 men will last 15 weeks. If 26 more men join them, for how many weeks will the provisions now be sufficient?

Ans ☐ 1. 14

☐ 2. 10

☒ 3. 12

☐ 4. 16

Question ID : 441009522917

Q.25 The slant height and the radius of a right circular cone are in the ratio of 5 : 3. If the volume of the cone is $4,710 \text{ cm}^3$, find its curved surface area. (Use $\pi = 3.14$).

- Ans
- ☒ 1. $1,177.5 \text{ cm}^2$
 - ☐ 2. $1,198.9 \text{ cm}^2$
 - ☐ 3. $1,248.6 \text{ cm}^2$
 - ☐ 4. $1,286.7 \text{ cm}^2$

Question ID : 4410091556635

Section : Computer knowledge

Q.26 Which menu in MS PowerPoint allows you to open an existing presentation?

- Ans
- ☐ 1. Insert
 - ☐ 2. Review
 - ☒ 3. File
 - ☐ 4. Design

Question ID : 4410092273236

Q.27 While preparing a multi-section annual report in a word processor, you need some pages to have landscape orientation for wider tables, while the rest remain in portrait. Which approach accurately ensures correct page setup without disrupting header and footer continuity?

- Ans
- ☐ 1. Change only the page margins or scale the tables to fit portrait orientation
 - ☒ 2. Insert section breaks before and after the landscape pages, then adjust the orientation for only those sections
 - ☐ 3. Use page breaks to separate each page and change the orientation individually
 - ☐ 4. Set page orientation for each page directly, regardless of section structure

Question ID : 4410092276857

Q.28 What is the primary role of the Control Unit in the Central Processing Unit?

- Ans
- ☐ 1. Stores program data
 - ☒ 2. Directs the execution of instructions
 - ☐ 3. Processes arithmetic calculations
 - ☐ 4. Manages hardware devices

Question ID : 4410092306929

Q.29 When preparing an Excel worksheet for printing, which option ensures that header rows appear at the top of every printed page?

- Ans
- ☒ 1. Print Titles option
 - ☐ 2. Print Area selection
 - ☐ 3. Row Height adjustment
 - ☐ 4. Page Setup dialog

Question ID : 4410092307263

Q.30 A user is sending an email with an important document but wants to avoid sending it without the required file. Which action should be performed before clicking the Send button?

- Ans
- ☐ 1. Type the file name in the message body before sending
 - ☐ 2. Check the file name in your computer folder before sending
 - ☒ 3. Verify the attachment is listed in the email draft before sending
 - ☐ 4. Upload the file and send immediately without checking attachment

Question ID : 4410092307611

Section : Electrical Engineering

Q.31 Which of the following best describes the function of an OR gate?

- Ans
- ☐ 1. It outputs 1 only if all inputs are 1
 - ☐ 2. It outputs 1 only if both inputs are the same
 - ☐ 3. It outputs 1 only if all inputs are 0
 - ☒ 4. It outputs 1 if at least one input is 1

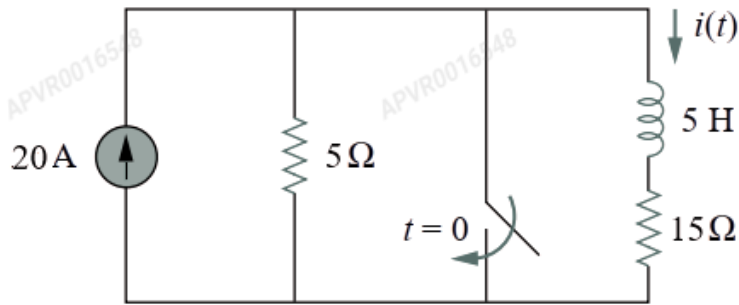
Question ID : 4410091805701

Q.32 A digital meter has a lowest full-scale range of 20 V and a resolution of 0.001. What is its sensitivity?

- Ans
- ☒ 1. 0.02 V
 - ☐ 2. 0.2 V
 - ☐ 3. 0.002 V
 - ☐ 4. 0.0002 V

Question ID : 4410091537564

Q.33 For the figure shown below, find the current $i(t)$ at $t = \infty$



- Ans
- ☒ 1. 20 A
 - ☒ 2. 0 A
 - ☒ 3. 15 A
 - ☒ 4. 5 A

Question ID : 4410091565518

Q.34 A designer wants to implement a function where the output is HIGH only if all four sensors in a system detect a signal producing logic 1. Which gate should be used?

- Ans
- ☒ 1. A four-input NAND gate
 - ☒ 2. A four-input AND gate
 - ☒ 3. A four-input OR gate
 - ☒ 4. A four-input NOR gate

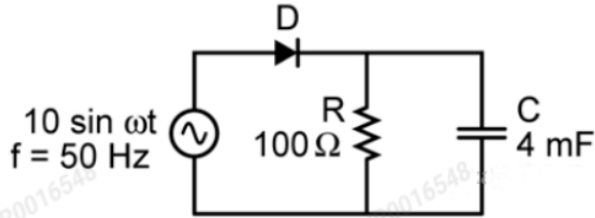
Question ID : 4410091792354

Q.35 A system with insufficient phase margin and high steady-state error is best compensated using _____.

- Ans
- ☒ 1. Rate-feedback compensator
 - ☒ 2. Lag compensator
 - ☒ 3. Lead compensator
 - ☒ 4. Lag-lead compensator

Question ID : 4410091576318

Q.36 The figure shows a half-wave rectifier. The diode D is ideal. The average steady-state current through the diode (in Amperes) is approximately _____.



- Ans
- ☒ 1. 1
 - ☒ 2. 10
 - ☒ 3. 0
 - ☒ 4. 0.1

Question ID : 4410091562742

Q.37 Which of the following options is correct for the negative clamper circuit?

- Ans
- ☒ 1. A clamper circuit is used to add an ac voltage to its input signal.
 - ☒ 2. Negative clamper circuit uses only diode and resistor.
 - ☒ 3. Negative clamper clamps the input signal waveform positively.
 - ☒ 4. A negative dc is added to the input signal in a negative clamper circuit.

Question ID : 4410091562540

Q.38 Which AC bridge is specifically employed for determining the relative permittivity (dielectric constant) of insulating materials?

- Ans
- ☒ 1. Schering Bridge
 - ☒ 2. Kelvin Bridge
 - ☒ 3. Wien Bridge
 - ☒ 4. Maxwell Bridge

Question ID : 4410091547326

Q.39 A cut-set in graph theory of electrical networks is defined as:

- Ans
- ☐ 1. A set of branches whose removal disconnects all nodes
 - ☒ 2. A minimum set of branches that, when removed, reduces the rank of the graph by one
 - ☐ 3. A set of all links in the graph
 - ☐ 4. A set of branches forming a closed loop

Question ID : 4410091558150

Q.40 Select the correct option regarding the following two statements labelled Assertion (A) and Reason (R).

Assertion (A): The economizer in a steam generating station is placed before the boiler in the flow path of flue gases.

Reason (R): The economizer preheats the feedwater using residual heat from the exhaust gases.

- Ans
- ☐ 1. Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - ☐ 2. (A) is true but (R) is false
 - ☐ 3. (A) is false but (R) is true
 - ☒ 4. Both (A) and (R) are true and (R) is the correct explanation of (A)

Question ID : 4410091572686

Q.41 Which of the following components in a nuclear reactor is used to slow down neutrons?

- Ans
- ☐ 1. Coolant
 - ☒ 2. Moderator
 - ☐ 3. Control rods
 - ☐ 4. Reactor core

Question ID : 4410091547023

Q.42 For a medium line, if shunt capacitance is neglected, the calculated voltage regulation will be _____.

- Ans
- ☐ 1. Independent of load power factor
 - ☐ 2. Lower than actual
 - ☒ 3. Higher than actual
 - ☐ 4. Equal to actual

Question ID : 4410091563918

Q.43 The output of a weighted resistor DAC is usually obtained using _____.

- Ans
- ☐ 1. A comparator
 - ☒ 2. A summing amplifier
 - ☐ 3. A rectifier
 - ☐ 4. A differentiator

Question ID : 4410091553886

Q.44 Dielectric heating is most suitable for heating which type of materials?

- Ans
- ☐ 1. Good electrical conductors
 - ☐ 2. Superconducting materials
 - ☐ 3. Metals with low resistivity
 - ☒ 4. Non-conducting materials such as plastics and wood

Question ID : 4410091562152

Q.45 Campbell's modification of the Heaviside bridge is used to:

- Ans
- ☐ 1. Eliminate detector sensitivity to phase errors
 - ☐ 2. Measure resistance using a known capacitance
 - ☐ 3. Measure mutual inductance using unknown self-inductance
 - ☒ 4. Measure self-inductance using a known mutual inductance

Question ID : 4410091537582

Q.46 For a first-order system with transfer function:

$$G(s) = \frac{1}{1 + Ts}$$

what is the phase angle $\phi(\omega)$ at the frequency $\omega=1/T$?

- Ans
- ☒ 1. -45°
 - ☐ 2. 0°
 - ☐ 3. -90°
 - ☐ 4. -180°

Question ID : 4410091553401

Q.47 During the turn-off process of SCR, the SCR requires:

- Ans
- ☐ 1. The negative voltage pulse should be given to the Gate
 - ☐ 2. The applied Gate current should be reversed
 - ☐ 3. Voltage across the device has to rise above rated value
 - ☒ 4. Current through the device to fall below holding current

Question ID : 4410091559720

Q.48 The fixed coil in an electrodynamic wattmeter is connected in series with the load and is known as the _____.

- Ans
- ☐ 1. Voltage Coil
 - ☒ 2. Current Coil
 - ☐ 3. Controlling Coil
 - ☐ 4. Potential Coil

Question ID : 4410091544752

Q.49 Consider a feedback control system with loop transfer function

$$G(s)H(s) = \frac{k(1 + 0.5s)}{s(1 + s)(1 + 2s)}$$

The type of the closed loop system is _____.

- Ans
- ☐ 1. Two
 - ☐ 2. Zero
 - ☒ 3. One
 - ☐ 4. Three

Question ID : 4410091660327

Q.50 A voltage quadruple is driven by an AC input. During steady-state operation, which one of the following interactions explains how the circuit maintains four times the peak input across the output despite only two conduction paths existing per cycle?

- Ans
- ☐ 1. The circuit uses reverse breakdown of diodes to boost voltage in discrete steps.
 - ☐ 2. Each capacitor doubles its voltage on every cycle until the quadruple level is reached.
 - ☒ 3. The diodes selectively route charge to different capacitors in alternating half-cycles, creating a cumulative stacking of capacitor voltages.
 - ☐ 4. Each conduction path simultaneously reinforces the charge on all capacitors every cycle.

Question ID : 4410091525532

Q.51 Damping in moving iron instruments is generally provided by:

- Ans
- ☐ 1. Magnetic damping
 - ☐ 2. Electrical damping
 - ☒ 3. Air friction damping
 - ☐ 4. Fluid (other than air) friction damping

Question ID : 4410091547760

Q.52 If cracks appear on the surface of a concrete foundation of a substation shortly after setting, which of the following is the most likely cause, considering the properties of concrete?

- Ans
- ☐ 1. Low permeability of concrete
 - ☐ 2. Excessive cement content
 - ☒ 3. Shrinkage due to rapid loss of moisture
 - ☐ 4. Slow setting of concrete

Question ID : 4410091804139

Q.53 A circuit compares two 4-bit numbers A and B. If $A = B$, the circuit must output $A + B$, otherwise it must output $A - B$. Which signal should be used as the MUX select line?

- Ans
- ☒ 1. $A = B$
 - ☐ 2. $A > B$
 - ☐ 3. $A < B$
 - ☐ 4. $(A > B) + (A < B)$

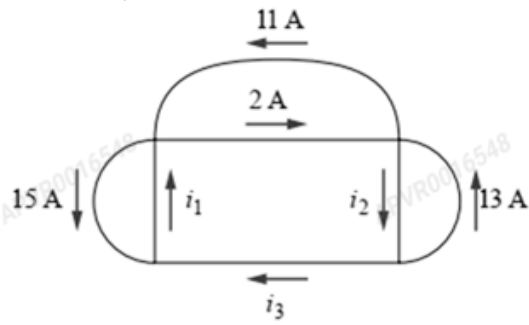
Question ID : 4410091560776

Q.54 Why are non-metallic conduits often preferred over steel conduits in residential buildings?

- Ans
- ☒ 1. Steel is a conductor (electrical) and prone to corrosion
 - ☐ 2. Steel is an insulator (electrical) and has low thermal expansion
 - ☐ 3. Steel is light-weight and has low strength
 - ☐ 4. Steel provides good thermal insulation

Question ID : 4410091787013

Q.55 For the figure below, find the values of currents, i_1 , i_2 , and i_3 . Every branch contains some element, do not consider them as short circuited branches.



Ans ☒ 1. $i_1 = 6 \text{ A}$, $i_2 = 4 \text{ A}$, $i_3 = -9 \text{ A}$

☒ 2. $i_1 = -6 \text{ A}$, $i_2 = 4 \text{ A}$, $i_3 = -9 \text{ A}$

☒ 3. $i_1 = 6 \text{ A}$, $i_2 = 4 \text{ A}$, $i_3 = 9 \text{ A}$

☒ 4. $i_1 = 6 \text{ A}$, $i_2 = -4 \text{ A}$, $i_3 = 9 \text{ A}$

Question ID : 4410091554835

Q.56 In an ideal diode-capacitor peak detector, which condition must be satisfied for the output capacitor to charge up to the peak value of the input waveform?

Ans ☒ 1. The diode must conduct continuously throughout the input cycle.

☒ 2. The diode must remain reverse-biased during the rising portion of the input signal.

☒ 3. The diode must conduct only when the input exceeds the capacitor voltage.

☒ 4. The capacitor must discharge rapidly between peaks to follow the input.

Question ID : 4410091525527

Q.57 According to resistance welding, from which source is the heat generated in percussion welding?

Ans ☒ 1. High-frequency induction

☒ 2. Intense electrical discharge for a very short time

☒ 3. Mechanical friction

☒ 4. Continuous arc heating

Question ID : 4410091558642

Q.58

Consider the transfer function $G_p(s) = \frac{ps^2 + 3ps - 2}{s^2 + (3+p)s + (2-p)}$ where 'p' is a positive real parameter. The maximum value of 'p' for which $G_p(s)$ remains stable is _____.

- Ans
- ☐ 1. 3
 - ☐ 2. 4
 - ☐ 3. 1
 - ☒ 4. 2

Question ID : 4410091661181

Q.59 For an inductor operating in DC steady state (i.e., $t \rightarrow \infty$), what is the equivalent circuit?

- Ans
- ☐ 1. A voltage source with value $L(di/dt)$
 - ☒ 2. A short circuit
 - ☐ 3. A resistor with value $R = L/\tau$
 - ☐ 4. An open circuit

Question ID : 4410091546075

Q.60 Which of the following sequence currents is ALWAYS zero during a line-to-line fault in an electrical power system?

- Ans
- ☐ 1. Negative sequence current
 - ☐ 2. Positive sequence current
 - ☒ 3. Zero sequence current
 - ☐ 4. None of the sequence currents are zero

Question ID : 4410091572697

Q.61 In an electrical network, the voltage across an unknown element is found as $325 \sin(314t - 45^\circ)$ volts while the current is found as $25 \sin(314t - 15^\circ)$ amperes. What possible circuit elements (Resistor (R), inductor (L), capacitor (C)) are present in the unknown element?

- Ans
- ☐ 1. R, L and C with $X_L > X_C$
 - ☐ 2. R and L only
 - ☐ 3. R only
 - ☒ 4. R, L and C with $X_L < X_C$

Question ID : 4410091552279

Q.62 In practical feedback control systems, a pure integral controller $G_c(s) = K_i/s$ is rarely used by itself. What is the main drawback that limits its standalone use?

- Ans
- ☒ 1. It's Bode plot has -20 dB/dec slope throughout
 - ☒ 2. It causes phase lag of -90° at all frequencies
 - ☒ 3. It makes the system type increase by exactly one
 - ☒ 4. It eliminates steady-state error for step input

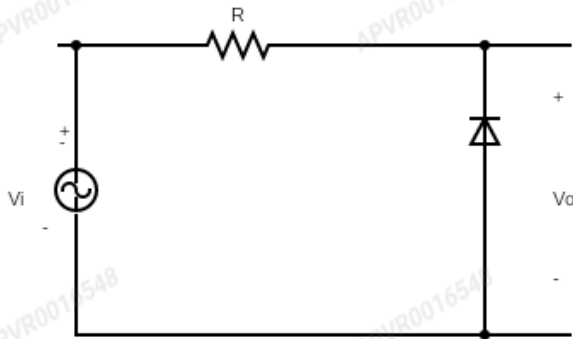
Question ID : 4410091576358

Q.63 Choose the correct statement for the given circuit. Assume diode to be an ideal diode.

S1: It is a shunt Positive clipper circuit.

S2: During positive half cycle, all the input positive half cycle voltage appears at the output.

S3: During negative half cycle, output voltage is zero.



- Ans
- ☒ 1. S1, S2 and S3
 - ☒ 2. Only S2 and S3.
 - ☒ 3. Only S2 and S1.
 - ☒ 4. Only S1 and S3.

Question ID : 4410091562544

Q.64 According to the magnitude rule of root locus, a point s lies on the root locus if _____.

- Ans
- ☒ 1. The real part of the complex variable s is negative
 - ☒ 2. The phase angle of the open-loop transfer function does not exist
 - ☒ 3. The magnitude of the closed-loop transfer function is unity
 - ☒ 4. The magnitude of the open-loop transfer function is unity

Question ID : 4410091579140

Q.65 Which of the following statements is TRUE regarding heat distribution inside a batch-type resistance furnace?

- Ans
- ☐ 1. Heat distribution depends only on radiation
 - ☐ 2. Radiation becomes the dominant method of heat distribution due to the circulating fan
 - ☐ 3. Conduction becomes the dominant method of heat distribution due to the circulating fan
 - ☒ 4. Convection becomes the dominant method of heat distribution due to the circulating fan

Question ID : 4410091555155

Q.66 What does SCADA stand for?

- Ans
- ☐ 1. Sequential Control and Data Analysis
 - ☒ 2. Supervisory Control and Data Acquisition
 - ☐ 3. Standard Control and Device Automation
 - ☐ 4. Systematic Coordination and Data Access

Question ID : 4410091805407

Q.67 A single-phase energy meter has meter constant of 1000 revolutions per unit and is tested for errors. The meter is connected to a load carrying 25 A current at 0.8 power factor with a voltage of 240 V applied across it for a period of 5 minutes. During this period, the number of revolutions made by the meter are 390. Which of the following is the error in the meter?

- Ans
- ☐ 1. 1.5% slower
 - ☐ 2. 3.5% faster
 - ☐ 3. 2.0% slower
 - ☒ 4. 2.5% slower

Question ID : 4410091533894

Q.68 Which of the following connections of the sequence network is involved in Line-To-Line (L-L) Fault?

- Ans
- ☐ 1. The positive and zero sequence networks connected in series.
 - ☐ 2. The positive, negative, and zero sequence networks connected in parallel.
 - ☐ 3. The positive, negative, and zero sequence networks connected in series.
 - ☒ 4. The positive and negative sequence networks connected in parallel.

Question ID : 4410091547036

Q.69 When comparing two concrete mixes for a transmission tower foundation, Mix A has a lower water-cement ratio than Mix B while maintaining adequate workability and compaction. What is the likely outcome for Mix A compared to Mix B? Note that the water cement ratio of both the mixes not less than 0.4.

- Ans
- ☒ 1. Mix A will have higher strength and durability
 - ☐ 2. Mix A will set faster
 - ☐ 3. Mix A will be more workable
 - ☐ 4. Mix A will have lower strength

Question ID : 4410091792932

Q.70 In three-phase power measurement, phase shifting is essential for accurate reactive power indication. Which arrangement is commonly used to obtain the required 90° phase shift using phase-shifting transformers?

- Ans
- ☐ 1. Two single-phase transformers connected in open-delta
 - ☐ 2. Three transformers connected in star-delta conversion
 - ☒ 3. Two autotransformers connected in V-V (open-delta) configuration
 - ☐ 4. Two autotransformers connected in Scott connection

Question ID : 4410091547404

Q.71 A step-down (buck) DC chopper is supplied from a stiff DC source of voltage V_s . The chopper operates by turning its main switch ON for a duration of T_{on} and OFF for a duration of T_{off} in each switching cycle. The load is such that current remains continuous during its operation. If the switching frequency is kept constant and the duty ratio of the chopper is increased, what will happen to the average output voltage across the load?

- Ans
- ☐ 1. It will vary independently of the duty ratio of the chopper.
 - ☐ 2. It will increase due to energy storage in the load.
 - ☒ 3. It will increase in direct proportion to the duty ratio of the chopper.
 - ☐ 4. It will remain fixed at the source voltage V_s of the chopper.

Question ID : 4410091563178

Q.72 Which of the following design modifications to a foundation in weak soil conditions for a substation helps to distribute load over greater area?

- Ans
- ☒ 1. Increasing the foundation base area
 - ☐ 2. Reducing the depth of the foundation
 - ☐ 3. Decreasing the base area to concentrate loads
 - ☐ 4. Using massive amounts of reinforcement

Question ID : 4410091813304

Q.73 Which alternator component accounts for the largest percentage of the overall losses at unity power factor and full load?

- Ans
- ☒ 1. Field Copper Loss
 - ☒ 2. Stray Load Loss
 - ☒ 3. Armature Copper Loss
 - ☒ 4. Friction and Windage Loss

Question ID : 4410091545574

Q.74 Why does projection welding allow multiple welds in a single operation?

- Ans
- ☒ 1. Because the weld pool is protected by inert gas
 - ☒ 2. Because alternating current distributes evenly across the workpiece
 - ☒ 3. Because electrodes are consumable and melt into the joint
 - ☒ 4. Because projections concentrate current at several points simultaneously

Question ID : 4410091557507

Q.75 It is referred to as _____ when a synchronous motor is utilized to control the voltage at the end of a transmission line by varying excitation.

- Ans
- ☒ 1. DC exciter
 - ☒ 2. Series reactor
 - ☒ 3. Induction generator
 - ☒ 4. Synchronous phase modifier

Question ID : 4410091545646

Q.76 Energy released per fission of U-235 is approximately:

- Ans
- ☒ 1. 200 MeV
 - ☒ 2. 2000 MeV
 - ☒ 3. 100 MeV
 - ☒ 4. 10 MeV

Question ID : 4410091558272

Q.77 In a static Kramer drive, speed control is limited to below synchronous speed primarily because:

- Ans
- ☒ 1. Slip power can flow only from rotor to supply
 - ☐ 2. The stator voltage cannot be controlled externally
 - ☐ 3. Commutation of the inverter restricts operation above synchronous speed
 - ☐ 4. The rotor resistance varies nonlinearly with load

Question ID : 4410091564230

Q.78 Which of the following statements about circuits with three-phase delta connections is true?

- Ans
- ☐ 1. Line currents are 60° apart
 - ☒ 2. Line voltage is equal to phase voltage
 - ☐ 3. Line voltage is $\sqrt{3}$ time of phase voltage
 - ☐ 4. Line current is equal to phase current

Question ID : 4410091545946

Q.79 Which significant advantage does the R-2R Ladder DAC architecture possess over the Binary-Weighted Resistor DAC architecture regarding the physical manufacturing of the circuit?

- Ans
- ☐ 1. It eliminates the need for a summing amplifier at the output stage.
 - ☐ 2. It allows the reference voltage to vary directly with the digital input code.
 - ☐ 3. It uses capacitors instead of resistors to perform the voltage division, reducing thermal noise.
 - ☒ 4. It requires only two specific precision resistor values regardless of the number of bits.

Question ID : 4410091538857

Q.80 A designer wants to implement a 2-input NOR gate using only 2-input NAND gates. How many minimum 2-input NAND gate is/are used to implement 2-input NOR gate?

- Ans
- ☐ 1. Five 2-input NAND gates
 - ☐ 2. Six 2-input NAND gates
 - ☐ 3. Eight 2-input NAND gates
 - ☒ 4. Four 2-input NAND gates

Question ID : 4410091792632

Q.81 A $4\frac{1}{2}$ digit digital voltmeter is used for voltage measurement. How the voltage value of 5.6279 V will be displayed on a 10 V range?

- Ans
- ☐ 1. 005.63 V
 - ☐ 2. 5.627 V
 - ☐ 3. 5.6279 V
 - ☒ 4. 05.627 V

Question ID : 4410091533405

Q.82 An SR latch starts in the reset state. For the input sequence $S = 1, R = 0 \rightarrow S = 1, R = 1 \rightarrow S = 0, R = 0$, determine the final output Q.

- Ans
- ☐ 1. Q = 0
 - ☐ 2. Q = 1
 - ☒ 3. Unpredictable or indeterminate
 - ☐ 4. High impedance

Question ID : 4410091553986

Q.83 What is range of the typical overall thermal efficiency of steam power plants?

- Ans
- ☐ 1. 60-70%
 - ☐ 2. 80-90%
 - ☒ 3. 30-40%
 - ☐ 4. 10-20%

Question ID : 4410091547028

Q.84 When designing a voltage control scheme for a single-phase inverter, why is internal modulation-based control preferred over varying the DC input voltage?

- Ans
- ☐ 1. It causes higher switching losses during normal inverter operation.
 - ☒ 2. It causes lower switching losses during normal inverter operation.
 - ☐ 3. It changes the inverter output frequency and phase more slowly (takes more time).
 - ☐ 4. It requires additional external power conditioning equipment.

Question ID : 4410091555264

Q.85 Which of the following is an advantage of using load-commutated chopper over other types of choppers?

- Ans
- ☐ 1. It uses a more complex and advanced commutation circuit than other types of choppers.
 - ☒ 2. It can operate at much higher frequencies than other types of choppers.
 - ☐ 3. It is capable of giving more output voltage than the input supply.
 - ☐ 4. It allows both voltage and current reversal through the load.

Question ID : 4410091563181

Q.86 If the rotor of a single-phase reluctance motor is perfectly cylindrical (no saliency), the reluctance torque _____.

- Ans
- ☐ 1. will be maximum
 - ☐ 2. depends on load
 - ☒ 3. will be zero
 - ☐ 4. will be half of the normal value

Question ID : 4410091563563

Q.87 A 200/100 V, 5 kVA two-winding transformer is reconnected as an autotransformer to deliver 200 V from a 300 V source. What will be the kVA rating of this autotransformer?

- Ans
- ☐ 1. 7.5 kVA
 - ☐ 2. 12.5 kVA
 - ☐ 3. 10 kVA
 - ☒ 4. 15 kVA

Question ID : 4410091549547

Q.88 In a DC shunt generator, the maximum efficiency occurs when:

- Ans
- ☒ 1. variable armature circuit loss = constant field winding loss + no load rotational loss
 - ☐ 2. variable field winding loss = constant armature circuit loss + no load rotational loss
 - ☐ 3. variable field winding loss + no load rotational loss = constant armature circuit loss
 - ☐ 4. variable armature circuit loss + no load rotational loss = constant field winding loss

Question ID : 4410091536607

Q.89 Which of the following characteristics is unique to a Star (Y) connection and not available in a standard three-wire Delta connection?

- Ans
- ☐ 1. Three-line conductors
 - ☐ 2. Suitability for high-current applications
 - ☐ 3. Equal line and phase voltages
 - ☒ 4. Provision of a neutral wire

Question ID : 4410091546016

Q.90 Earthing in electrical installations of buildings is primarily provided to:

- Ans
- ☒ 1. Protect people from electric shock by safely dissipating fault current
 - ☐ 2. Increase the voltage supplied to electrical appliances in buildings
 - ☐ 3. Improve the efficiency of the electrical system in buildings
 - ☐ 4. Reduce the overall electricity consumption of appliances

Question ID : 4410091787015

Q.91 A reactor is connected between the two converters in a three-phase dual converter operating in circulating current mode. What is the primary purpose of this reactor?

- Ans
- ☐ 1. Improve the input power factor.
 - ☐ 2. Limit the load current ripple.
 - ☐ 3. Minimize the effect of source inductance.
 - ☒ 4. Limit circulating current between the two converters

Question ID : 4410091550320

Q.92 In a connected network graph, the number of fundamental cut-sets is equal to:

- Ans
- ☐ 1. Number of loops
 - ☐ 2. Number of chords
 - ☐ 3. Total number of nodes
 - ☒ 4. Number of twigs

Question ID : 4410091558155

Q.93 The main components of a single-phase induction motor generally refer to:

- Ans ☒ 1. Stator and Rotor with auxiliary winding
☐ 2. Stator only
☐ 3. Armature and field winding
☐ 4. Rotor only

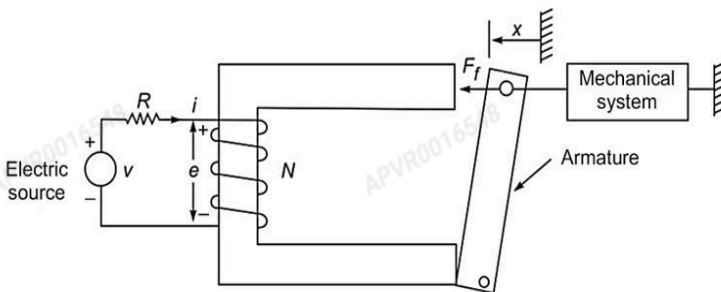
Question ID : 4410091554922

Q.94 In a two-phase AC servomotor operating under normal closed-loop conditions, the developed torque becomes zero when ____

- Ans ☒ 1. The error signal applied to the control winding becomes zero
☐ 2. The phase displacement between the reference winding and control winding currents becomes zero
☐ 3. The magnitude of the reference winding current remains constant
☐ 4. The rotor reaches synchronous speed

Question ID : 4410091549503

Q.95 Consider the single-excited magnetic system shown in the figure. The exciting coil of N turns carries a constant current i , and the armature is free to move in the horizontal direction (x) under the electromagnetic force F_f (as shown in the figure). Which of the following changes in the system will cause an increase in the inductance of the exciting coil?



- Ans ☐ 1. Decrease in the permeability of the magnetic core material due to its saturation
☐ 2. Increase in the air gap length due to movement of the armature away from the fixed core
☒ 3. Decrease in the air gap length due to movement of the armature towards the fixed core
☐ 4. Increase in the total magnetic circuit reluctance due to decrease in air gap length

Question ID : 4410091545941

Q.96 A three-phase semi-converter operates as a six-pulse converter under which condition?

- Ans
- ☒ 1. When the firing angle α is less than 60°
 - ☐ 2. When the firing angle α is equal to 120°
 - ☐ 3. When the firing angle α is greater than 90°
 - ☐ 4. When the firing angle α is greater than 60°

Question ID : 4410091564128

Q.97 In a Constant Frequency System of Time Ratio Control in chopper operation, how is the average output voltage varied?

- Ans
- ☐ 1. By keeping the ON-time (T_{ON}) constant and varying the OFF-time (T_{OFF})
 - ☒ 2. By varying the ON-time (T_{ON}) while keeping the total time period (T) constant
 - ☐ 3. By varying both T_{ON} and T_{OFF} while keeping their ratio constant
 - ☐ 4. By varying the total time period (T) while keeping T_{ON} (ON Time) constant

Question ID : 4410091550660

Q.98 A designer needs a gate that outputs HIGH only when both inputs are identical, regardless of whether they are LOW or HIGH. Which gate should be chosen?

- Ans
- ☐ 1. NAND
 - ☒ 2. XNOR
 - ☐ 3. AND
 - ☐ 4. NOR

Question ID : 4410091792326

Q.99 Which modification in JK flip-flop design removes the possibility of race-around condition even when $J = K = 1$?

- Ans
- ☒ 1. Using edge triggering
 - ☐ 2. Using NAND gates at outputs
 - ☐ 3. Reducing clock frequency
 - ☐ 4. Increasing propagation delay

Question ID : 4410091576283

Q.100 A hydro power plant has a head of 50 meters and water is discharged at a rate of 600 cubic meters per second for electricity generation. It generates an output power of 255 megawatts. What is the efficiency of the plant?

{Note: Take acceleration due to gravity, $g = 10 \text{ m/s}^2$ }

Ans ☒ 1. 95%

☒ 2. 85%

☒ 3. 65%

☒ 4. 75%

Question ID : 4410091572681